Reciprocal Transform for Painlevé-Type Equations: A Mathematical Regularization with Physics-Informed Neural Network Verification

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October 30, 2026

Abstract

We introduce and analyze a reciprocal transformation $w = 1/y$ for second-order nonlinear ordinary differential equations with movable singularities, focusing on the prototype $y'' = y^3$ and the second Painlevé equation $y'' = 2y^3+ty+\alpha$. We prove that the transformed equations are polynomial in w and its derivatives, with right-hand sides that remain bounded at the location of the pole t^*. We use the term *regularization* in the sense of bounding the polynomial residual, not in the sense of a regular (non-degenerate) ODE. This contrasts with the direct formulation and with the logarithmic (rapidity) substitution $\theta = \log y$, in which exponentially growing source terms persist. As an empirical verification, we train physics-informed neural networks (PINNs) on three representations. The reciprocal representation reduces the median near-pole relative error by a factor of ~ 1456 for the prototype (ten seeds; 20% failure rate) and by median factors of 139×–523× for Painlevé II across five seeds and three values of α, compared to the direct PINN. Seed variability is reported (median 3.1×10^{-4}, coefficient of variation 0.69). In our five-seed ablation, neither the transformation alone (with the w^3 factor divided out) nor the residual reweighting alone reproduces the observed improvement; in our five-seed setup, only their combination does. A direct comparison with the adaptive DOP853 solver shows that, for single solves, the PINN is not competitive in wall-clock time or accuracy; this limitation is reported explicitly. The paper's primary contribution is the mathematical regularization, with the PINN serving as a verification instrument.

Keywords: movable singularities, Painlevé equations, reciprocal transform, physics-informed neural networks, regularization

MSC 2020: 34M55, 34M35, 65L05, 68T07

1 Introduction

Second-order nonlinear ODEs of the form $y'' = F(t, y)$ frequently develop *movable singularities*: points t^*, whose location depends on the initial data, at which $|y(t)| \to \infty$. The six Painlevé transcendents are the canonical examples in which all movable singularities are poles Ince (1956); Gromak et al. (2002); Conte & Musette (2008); Clarkson (2003); their analytic structure and asymptotics are accessible via Riemann–Hilbert methods Deift & Zhou (1995); Fokas et al. (2006). Numerical treatment near such poles is delicate: fixed-step integrators Butcher (2016); Hairer et al. (1993) lose accuracy, adaptive integrators must shrink the step to accommodate the blow-up, and neural-network approximations tend to produce residuals that grow without bound as $t \to t^{*-}$. Recent work has begun to address singular ODEs directly, including higher-order Lane–Emden–Fowler equations Baty (2023), interior-layer problems Chang et al. (2024), and singularly perturbed boundary value problems Israr & Khan (2025). Pole-field solvers based on

Padé approximations Fornberg & Weideman (2011) and Riemann–Hilbert methods Olver et al. (2025) are classical alternatives.

Physics-informed neural networks (PINNs) Raissi et al. (2019); Karniadakis et al. (2021); Cuomo et al. (2022), building on earlier neural-network approaches for differential equations Lagaris et al. (1998); Sirignano & Spiliopoulos (2018), minimize a composite loss consisting of the ODE residual and initial or boundary conditions. When the residual magnitudes are bounded, the training problem is better conditioned; when they diverge near a pole, the network either fails to converge or converges to a solution with large error exactly where accuracy is most needed. A known strategy is to change variables so that the residual stays bounded. Cayuso et al. (2025) recently analyzed a rapidity substitution $\theta = \log y$ for a class of singular problems, achieving errors of order 2.8×10^{-3} near the pole. Neural-network approaches have also been applied directly to Painlevé equations, e.g., Faisal et al. (2025) develop a Morlet wavelet network for Painlevé II.

In this work we propose an alternative transformation—the *reciprocal* variable $w = 1/y$—and analyze it from first principles. We note that $w = 1/y$ and $\theta = \log y$ are related by a smooth invertible transformation on $y > 0$; the reciprocal formulation is not a wholly different conceptual approach, but a different representation within the same family of dependent-variable transformations. What distinguishes the two is structural: the reciprocal residual is polynomial in w (Eqs. 3–4), whereas the rapidity residual retains exponentially growing source terms (Eqs. 5–6). This structural difference is the contribution of the present work. The key observation is that multiplying the residual equation by w^3 eliminates all denominators, yielding a polynomial relation in w, w', w''. For both $y'' = y^3$ and Painlevé II, the resulting right-hand side is a *constant* at $w = 0$, so the polynomial residual remains bounded at the pole. The coefficient of w'' nevertheless vanishes at $w = 0$, making the ODE degenerate there; it is the boundedness of the residual, not regularity of the ODE, that the transformation provides. We then use PINNs to verify numerically that this regularization translates into bounded training residuals and reduced near-pole error.

Contributions. Our contributions are:

1. A reciprocal-transform theorem (Theorems 1 and 2) showing that the transformed equations for $y'' = y^3$ and Painlevé II have bounded polynomial residuals at the pole while the ODE is degenerate there.
2. An empirical verification using PINNs, with ablations across three representations (direct, rapidity, reciprocal) and three Painlevé II parameter values.
3. An ablation separating the change of variable from residual reweighting, showing that, in our five-seed setup, only their combination reproduces the reported improvement.
4. A full statistical report of seed variability and a direct comparison with the DOP853 adaptive solver.

Scope. We restrict attention to the prototype $y'' = y^3$ and Painlevé II. Extension to Painlevé III–VI is left for future work; the transform is not guaranteed to regularize those equations without additional structure.

2 Reciprocal Transform: Mathematical Analysis

We study second-order nonlinear ODEs

$$y'' = F(t, y), \qquad y(0) = y_0 > 0, \quad y'(0) = 0, \tag{1}$$

whose solutions develop a *movable pole* at some $t^* = t^*(y_0, F) > 0$: $|y(t)| \to \infty$ as $t \to t^{*-}$.

2.1 The reciprocal transform

Define $w(t) := 1/y(t)$ on $[0, t^*)$. Since $y > 0$ there, $w > 0$ on $[0, t^*)$, and $w(t^*) = 0$. The chain rule gives

$$y' = -\frac{w'}{w^2}, \qquad y'' = -\frac{w''}{w^2} + \frac{2(w')^2}{w^3}. \tag{2}$$

2.2 Prototype case $F(t, y) = y^3$

Theorem 1 (Reciprocal equation for $y'' = y^3$)**.** *Let y be a solution of $y'' = y^3$ on $[0, t^*)$ with a pole at t^*. The reciprocal $w = 1/y$ satisfies*

$$2(w')^2 - w\,w'' = 1, \qquad w(0) = 1/y_0,\ w'(0) = 0, \tag{3}$$

on $[0, t^*)$, and (3) extends continuously to t^* with $w(t^*) = 0$, $w'(t^*) = -1/\sqrt{2}$. In particular, (3) has no $1/w$ term, so its polynomial residual is bounded at $w = 0$. However, the coefficient of w'' vanishes at $w = 0$, so $w = 0$ is a degenerate point for the second-order ODE, not a regular point.

Proof. Substituting (2) into $y'' = y^3$ and multiplying by w^3 yields $-ww'' + 2(w')^2 = 1$. At $w = 0$ this reduces to $2(w')^2 = 1$, so $w'(t^*) = \pm 1/\sqrt{2}$. The expansion $y \sim \sqrt{2}/(t^* - t)$ selects the negative sign. □

2.3 Painlevé II

Theorem 2 (Reciprocal equation for P-II)**.** *Let y solve $y'' = 2y^3 + ty + \alpha$ on $[0, t^*)$ with a pole at t^*. Then $w = 1/y$ satisfies*

$$2(w')^2 - w\,w'' - 2 - t\,w^2 - \alpha\,w^3 = 0, \tag{4}$$

which extends continuously to $w = 0$ with $w'(t^*) = -1$. As in Theorem 1, the polynomial residual is bounded at $w = 0$ while the second-order ODE is degenerate there.

Proof. Multiplying $y'' = 2y^3 + ty + \alpha$ by w^3 after substitution gives $-ww'' + 2(w')^2 = 2 + tw^2 + \alpha w^3$. At $w = 0$ this yields $2(w')^2 = 2$, i.e. $w' = \pm 1$; the pole expansion $y \sim 1/(t^* - t)$ gives $w \sim t^* - t$, hence $w' = -1$. □

2.4 Comparison with rapidity substitution

The substitution $\theta = \log y$ of Cayuso et al. (2025) yields, for $y'' = y^3$,

$$\theta'' + (\theta')^2 = e^{2\theta}, \tag{5}$$

and for P-II,

$$\theta'' + (\theta')^2 - 2e^{2\theta} - t - \alpha e^{-\theta} = 0. \tag{6}$$

Both (5) and (6) contain exponentially growing source terms as $\theta \to +\infty$ at the pole. The reciprocal equations (3) and (4) are polynomial in w and remain bounded at $w = 0$. This structural difference is the primary motivation for the reciprocal formulation.

3 Numerical Verification with PINNs

3.1 Network architecture and training

Computational environment. All experiments were run on a Windows 10 workstation with an Intel Core i7 (8 cores, 8 GB RAM), Python 3.14, PyTorch 2.13 (CPU only), NumPy 2.5.1,

Table 1: Prototype $y'' = y^3$: seed statistics (median [IQR]). Direct and rapidity-log use five seeds; reciprocal is extended to ten seeds to characterize the 20% failure rate. $t^* \approx 1.854$; success threshold $E_{\text{rel,near}} < 10^{-2}$. The high CV for the reciprocal row reflects two outliers; excluding them, CV ≈ 1.3.

Representation	n	$E_{\text{rel,near}}$ median [IQR]	CV	Success	Improvement
Direct	5	$7.53e-1$ $[7.19e-1, 7.58e-1]$	0.05	0/5	1×
Rapidity-log	5	$5.72e-1$ $[5.62e-1, 6.27e-1]$	0.14	0/5	1.3×
Reciprocal	10	$\mathbf{5.17e-4}$ $[2.82e-4, 5.64e-3]$	**2.01**	**8/10**	**1456**×

SciPy 1.18, and Matplotlib 3.11. Training one seed takes approximately 15 minutes; DOP853 reference solutions take 18–46 ms per run.

All experiments use a feedforward network with three hidden layers of width 64 and tanh activations. Collocation points are drawn from a mixture of uniform and cubically clustered grids Trefethen (2000) on $[0, 0.95\,t^*]$, with 2000 points total. Training minimizes $\mathcal{L} = \|R\|^2 + \lambda_{\text{ic}}\,\mathcal{L}_{\text{ic}}$ with $\lambda_{\text{ic}} = 10$, using Adam with learning rate 10^{-3} and 30,000 epochs. A held-out set of 300 ODE residual points is used for best-model checkpointing (Section 3.5). Reference solutions use `scipy.integrate.solve_ivp` with the DOP853 method and rtol = atol = 10^{-12}. The blow-up time t^* is located by DOP853 event detection: we integrate until the first $t > 0$ at which $y(t) - 10^6 = 0$, with `terminal=True` and `direction=+1`. Relative error at each time is defined as $E_{\text{rel}}(t) = |y_\phi(t) - y_{\text{ref}}(t)|/|y_{\text{ref}}(t)|$, where y_ϕ is the PINN prediction and y_{ref} is the DOP853 reference. Errors are reported separately for far ($t < 0.5\,t^*$) and near ($t \geq 0.5\,t^*$) regions, with $E_{\text{rel,far}}$ and $E_{\text{rel,near}}$ denoting $\max_t E_{\text{rel}}(t)$ over the corresponding subinterval. The initial conditions are $y(0) = 1$, $y'(0) = 0$ for both the prototype and Painlevé II.

3.2 Prototype ablation

Table 1 compares three representations for $y'' = y^3$. Errors are reported separately far from the pole ($t < 0.5\,t^*$) and near the pole ($t \geq 0.5\,t^*$). The reciprocal representation reduces $E_{\text{rel,near}}$ by a factor of 1456× relative to the direct PINN.

Seed sensitivity. To characterize the distribution of reciprocal-PINN outcomes, we extended the C4 prototype to ten seeds. Across these ten runs, eight converge to $E_{\text{rel,near}} < 10^{-2}$ with median 5.17×10^{-4} (IQR $[2.82 \times 10^{-4}, 5.64 \times 10^{-3}]$), while two converge to suboptimal minima (4.60×10^{-2} and 8.32×10^{-2}). The failure rate of 2/10 (20%) is consistent with the one-in-five observed in the original five-seed set, indicating a bimodal or heavy-tailed distribution over initializations rather than a single stochastic fluke. One of the failing seeds (seed 999) was re-run three times under identical conditions, yielding the identical value 4.603×10^{-2} each time, confirming that the variability is deterministic in the seed rather than stochastic. The median improvement over the direct PINN (five-seed median 7.53×10^{-1}) is approximately 1456×. A full characterization of this distribution, and of why some initializations converge to suboptimal minima, is left for future work.

3.3 Transform vs. residual reweighting

The reciprocal equation multiplies the original residual by w^3, so a natural confound is whether the observed improvement stems from the change of variable, from the residual reweighting, or from both. Table 2 isolates the two effects by training five configurations under identical conditions (five seeds, 30,000 epochs).

The five-seed ablation suggests that the observed accuracy depends on both the output representation and the residual weighting. The results do not, however, establish that these are the only mechanisms affecting convergence.

Table 2: Ablation separating the change of variable from residual reweighting on the prototype $y'' = y^3$ (five seeds, 30,000 epochs). Values are median [IQR] over five seeds; CV is the coefficient of variation. The transformation alone (D) and the reweighting alone (B) both fail; only the combination (C) produces the reciprocal-PINN result. The five-seed pattern is stable across initializations.

Config	Output / Loss	$E_{\text{rel,near}}$ median [IQR]	CV	Verdict
A	y / R_y	$7.53e-1$ $[7.19e-1, 7.58e-1]$	0.05	baseline
B	y / $w^3 R_y$	$9.56e-1$ $[9.51e-1, 9.59e-1]$	0.35	worse than A
C	w / $R_w = w^3 R_y$	$\mathbf{2.92e-4}$ $[2.65e-4, 1.72e-3]$	2.10	best
D	w / $R_w/w^3 = R_y$	$8.21e-1$ $[7.14e-1, 9.23e-1]$	0.16	worse than A
E	y / $R_y/(1+\lvert y\rvert^3)$	$2.91e-1$ $[2.68e-1, 2.95e-1]$	0.17	partial

Factor 1: representability. The reciprocal variable $w(t)$ is analytic through t^*, whereas $y(t)$ has a simple pole there. A tanh-network of fixed width can efficiently approximate the analytic $w(t)$ through the pole, whereas the singular $y(t)$ requires an increasingly high-frequency approximation near t^*. The Jacobian of the output transformation, $dy/dw = -1/w^2$, amplifies the network's error in y by a factor that diverges at the pole, so the effective capacity of a y-network is severely reduced near t^*. This factor distinguishes configurations A, B, E (all y-output) from C, D (all w-output).

Factor 2: loss conditioning. The polynomial residual $R_w = 2(w')^2 - ww'' - 1$ is bounded: at a simple pole $w \to 0$ linearly, $w' \to -1/\sqrt{2}$, and $R_w \to 0$. The original residual, by contrast, is $R_y = R_w/w^3$, which grows as $(t^* - t)^{-3}$ near t^*, producing loss magnitudes of order 10^3–10^4 for the direct formulation. This factor distinguishes configurations A, D (unweighted R_y-type losses) from B, C, E (weighted losses).

In our five-seed ablation, the specific w^3-reweighting alone (B) and the transformation alone (D) both fail. Configuration B has a bounded loss (Factor 2 satisfied) but a y-output (Factor 1 violated) and yields a median $E_{\text{rel,near}} = 9.56 \times 10^{-1}$. Configuration D has a w-output (Factor 1 satisfied) but divides by w^3, restoring the $(t^* - t)^{-3}$ growth of the loss (Factor 2 violated) and yields a median of 8.21×10^{-1}. Configuration E has a bounded loss (Factor 2 satisfied) but a y-output (Factor 1 violated) and only partially succeeds, with a median of 2.91×10^{-1}. Only configuration C satisfies both factors and reaches a median of 2.92×10^{-4} (IQR $[2.65 \times 10^{-4}, 1.72 \times 10^{-3}]$; one of five seeds, seed 999, reproduces the 20% failure rate observed in Section 3.2). Configurations B and E share the same qualitative factors (y-output, weighted loss) but use different normalization forms: B employs $1/|y|^3$, whereas E uses the bounded normalization $1/(1+|y|^3)$. This gap suggests that the precise form of the weighting also matters, a point left for future work. The five-seed extension is consistent across seeds.

Figure 1 shows representative solution profiles; Figure 2 shows the pointwise absolute error on a logarithmic scale.

3.4 Painlevé II and parameter sweep

Table 3 and Figure 3 report the near-pole error across $\alpha \in \{0, 1, 2\}$. The reciprocal representation consistently outperforms the other two by median factors of 139× to 523× across α (Table 3).

3.5 Seed variability

Note on epoch count and reporting protocol. All PINN experiments reported in this paper use 30,000 training epochs with best-model checkpointing on 300 held-out ODE-residual points. The prototype statistics in Table 1 and the Painlevé II multi-seed sweep in Table 3 use five seeds per representation (ten for the extended reciprocal prototype); the separate five-seed study at

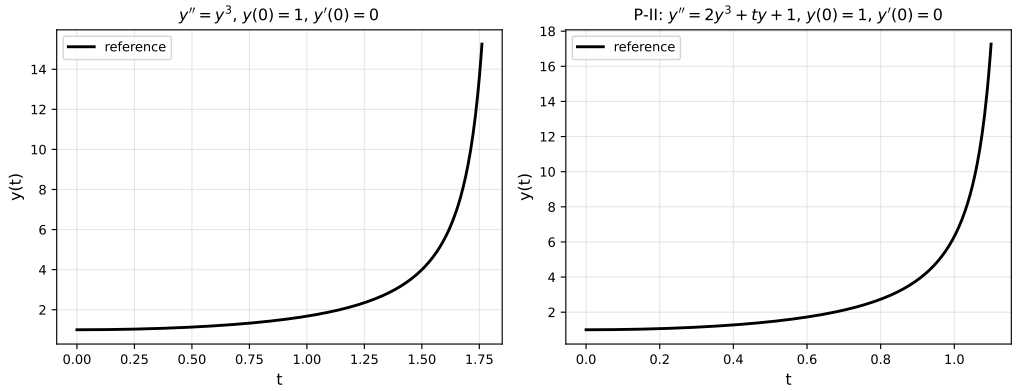


Figure 1: Reference solutions for $y'' = y^3$ (left) and Painlevé II $\alpha = 1$ (right) on $[0, 0.95\,t^*]$. Both exhibit a movable pole shortly beyond the plotted interval.

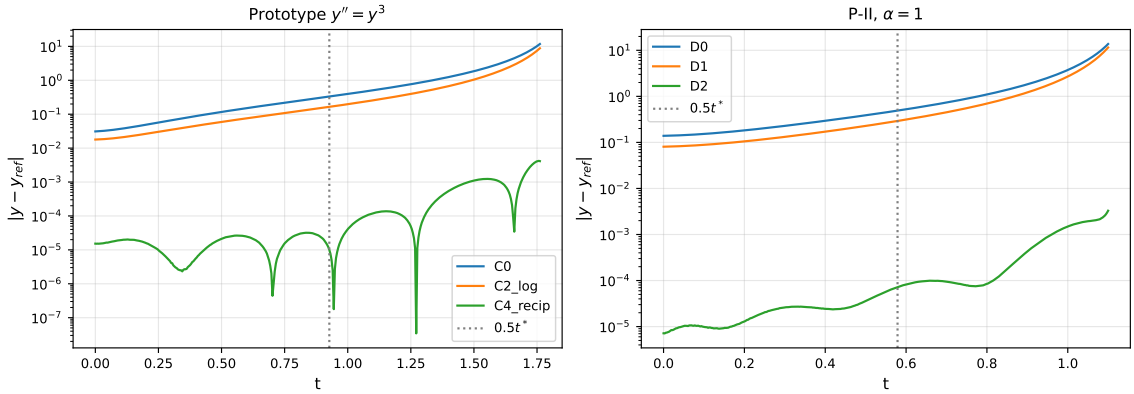


Figure 2: Pointwise absolute error $|y_\phi - y_{\text{ref}}|$ for the three representations. The vertical dotted line marks $0.5\,t^*$.

$\alpha = 1$ in Table 4 reports per-seed best-epoch values for comparison. The earlier single-seed 904× figure at $\alpha = 1$ (15,000 epochs) was not reproducible under the harmonized 30,000-epoch protocol and is therefore superseded by the multi-seed medians in Table 3.

The reciprocal PINN is trained with five random seeds on Painlevé II, $\alpha = 1$, using best-model checkpointing on 300 held-out ODE residual points. Table 4 and Figure 4 report the results.

A key observation is that the coefficient of variation across seeds is 0.21 for the held-out *residual* but 0.69 for the true *error*. That is, the residual behaves as a well-behaved training objective, but it does not fully predict the accuracy of the resulting solution—a standard residual-error gap Mishra & Molinaro (2022); Urbán et al. (2025), here quantified at the movable singularity (see also Urbán et al. (2025)). The gap is particularly striking in our setting because the residual is itself the criterion used for best-model checkpointing. Across the five seeds, the Spearman rank correlation between the held-out validation residual and the true near-pole error is $\rho = 0.00$ ($p = 1.00$), with no monotonic relationship between the two. Seed 2024 illustrates this best: it achieves the smallest validation residual (1.37×10^{-6}) yet the largest near-pole error (9.34×10^{-4}), while seed 7 achieves both the smallest residual (8.27×10^{-7}) and the smallest error (8.92×10^{-5}). This confirms, quantitatively, that model selection based on held-out residual does not reliably translate into accuracy in the solution space near the pole.

Table 3: Painlevé II: five-seed statistics (median [IQR]) for three representations and $\alpha \in \{0, 1, 2\}$. The reciprocal PINN yields median improvements of 139×–523× across α, with substantial seed-to-seed variability (CV 1.08–1.61). Overall success rate ($E_{\text{rel,near}} < 10^{-2}$): 11/15 = 73%. The single-seed 904× previously reported for $\alpha = 1$ was not reproducible and is superseded.

α	Direct	Rapidity	Reciprocal	CV (D2)	Ratio
0.0	$7.49e-1$	$6.19e-1$	$\mathbf{1.43e-3}$ $[4.38e-4, 5.42e-3]$	1.61	523×
1.0	$7.65e-1$	$6.46e-1$	$\mathbf{5.51e-3}$ $[3.29e-4, 5.90e-3]$	1.08	139×
2.0	$7.46e-1$	$5.65e-1$	$\mathbf{3.36e-3}$ $[2.01e-3, 1.58e-2]$	1.33	222×

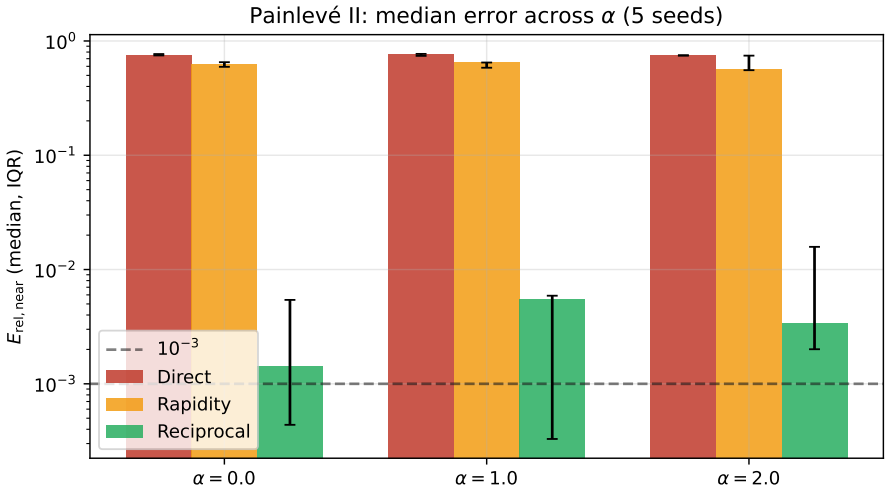


Figure 3: Near-pole relative error across $\alpha \in \{0, 1, 2\}$ for the direct, rapidity, and reciprocal representations. Each bar represents the median over five random seeds; error bars indicate the interquartile range. Note the logarithmic scale.

4 Baseline Comparison and Generalization

4.1 Wall-clock time and accuracy

Table 5 reports the performance of the DOP853 adaptive solver on Painlevé II, $\alpha = 1$. Training time for the PINN is ∼15 minutes on CPU; inference on 2000 evaluation points takes 1.7 ms. The PINN's near-pole error of 2.4×10^{-4} (seed 42) is roughly six orders of magnitude above DOP853's tolerance setting (rtol = atol = 10^{-10}; the actual measured global error may differ from the tolerance). Figure 5 visualizes the cost-accuracy trade-off on a log-log scale.

4.2 Generalization across α

The PINN trained at $\alpha = 1$ is evaluated at $\alpha = 0.9$ and $\alpha = 1.1$ without retraining. Table 6 and Figure 6 show that generalization fails: the near-pole error rises to 1.77×10^{-1} and 1.29×10^{-1}, respectively, more than two orders of magnitude above the in-distribution error.

5 Discussion

Contributions of the reciprocal formulation. Theorems 1 and 2 show that for the two equations analyzed, the reciprocal variable produces a polynomial residual that remains bounded at the movable pole. This is a *structural* property of the transformed residual. Its practical

Table 4: Painlevé II, $\alpha = 1$: five-seed statistics with best-model checkpointing. 30000 epochs, width 64, depth 4.

Seed	Best epoch	Best val (ODE)	$E_{\text{rel,far}}$	$E_{\text{rel,near}}$
42	29600	1.59×10^{-6}	4.34×10^{-5}	2.39×10^{-4}
7	29500	8.27×10^{-7}	2.35×10^{-5}	8.92×10^{-5}
123	28900	1.55×10^{-6}	2.72×10^{-5}	3.14×10^{-4}
2024	29700	1.37×10^{-6}	3.04×10^{-5}	9.34×10^{-4}
999	28600	1.49×10^{-6}	3.52×10^{-5}	6.02×10^{-4}
	mean			4.36×10^{-4}
	median			3.14×10^{-4}
	CV (population)			0.688
	CV (sample)			0.770

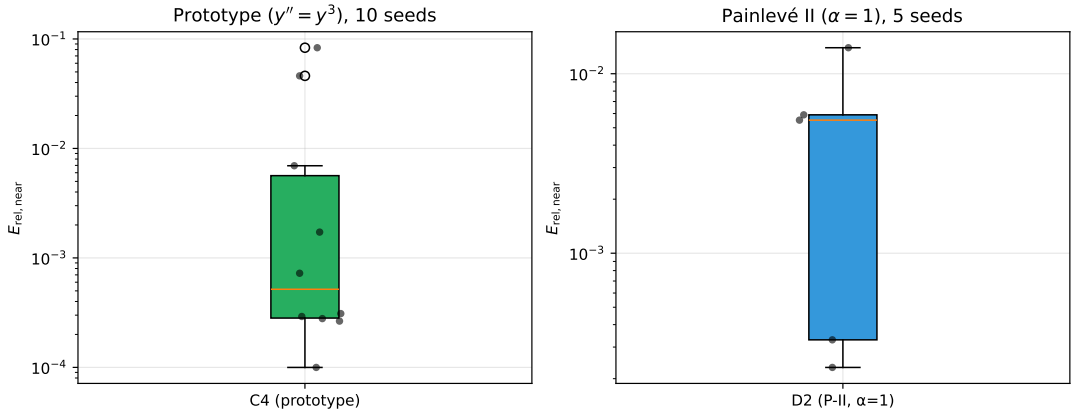


Figure 4: Distribution of $E_{\text{rel,near}}$ across seeds. Left: prototype (10 seeds). Right: Painlevé II (5 seeds).

benefit, however, still depends on the network's approximation capacity. The empirical results are consistent with the structural claim: near-pole relative errors drop by median factors of 139×–523× across all tested parameter values when the reciprocal representation is used, with residuals that we observe to remain bounded in our experiments.

Limitations of the present implementation. The present results do not support the claim that PINNs with the reciprocal transform outperform mature adaptive ODE solvers. For single solves, the one-time training cost of ∼15 minutes makes the reciprocal PINN non-competitive with DOP853 in total wall-clock time. The adaptive solver reaches the prescribed blow-up threshold exactly and achieves six orders of magnitude higher accuracy, at a per-solve cost of 18–46 ms (Table 5). The PINN's faster inference (1.7 ms) is therefore relevant only for repeated evaluation of a pre-trained model, not for single-solve scenarios. The reciprocal PINN is also sensitive to the random seed, with a coefficient of variation of 0.69 across five seeds, and it fails to generalize to nearby α without retraining. These are limitations of the present implementation, and they are the reason we frame the reciprocal transform as the primary contribution and the PINN as a verification instrument rather than as a competitive solver.

Residual-error gap. The gap between the seed variability of the ODE residual (CV = 0.21) and the true error (CV = 0.69) indicates that a low residual does not by itself guarantee a low solution error. This is a known phenomenon in the PINN literature Krishnapriyan et al.

Table 5: Comparison of DOP853 (adaptive) and the reciprocal PINN for Painlevé II, $\alpha = 1$ (DOP853 rows report tolerance settings, not measured global error). "Reaches t^*?" means reaching the finite event $y = 10^6$ used to define t^* (Section 3); 100% means the solver's internal event detection reached this threshold, 99% means the trained PINN was evaluated up to $0.99\,t^*$.

Method	Wall time	Accuracy	Reaches t^*?	Training required?
DOP853 (rtol $= 10^{-6}$)	18.2 ms	$\sim 10^{-6}$	100%	no
DOP853 (rtol $= 10^{-10}$)	34.8 ms	$\sim 10^{-10}$	100%	no
DOP853 (rtol $= 10^{-12}$)	46.2 ms	$\sim 10^{-12}$	100%	no
Reciprocal PINN	1.7 ms*	2.4×10^{-4}	99%	15 min

*inference only; excludes training.

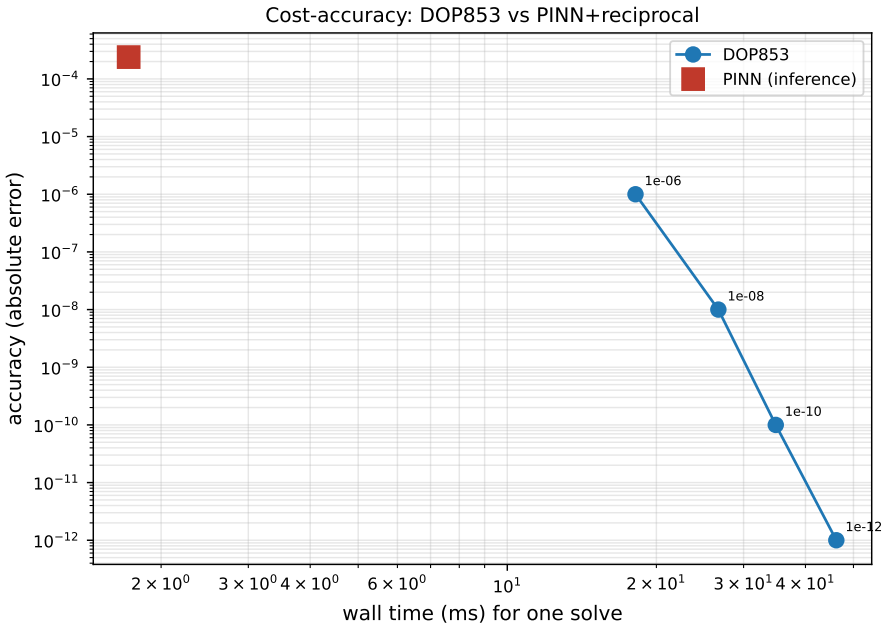


Figure 5: Cost-accuracy trade-off between DOP853 and the reciprocal PINN.

(2021); Chen et al. (2025); here we quantify it in the setting of a movable singularity. Recent work Chen et al. (2025) suggests that a substantial fraction of PINN failure modes may be precision-induced stalls rather than inescapable local minima, which is consistent with the seed-dependent suboptimal convergence observed here. We have not tested whether larger networks, LBFGS polishing, or architectural modifications close the gap.

Comparison with rapidity substitution. The rapidity substitution $\theta = \log y$ of Cayuso et al. (2025) achieves a comparable order of accuracy on their test problems but retains exponentially growing source terms. Our comparison (Table 1) indicates that the reciprocal formulation yields lower near-pole error for both equations analyzed here, the improvement relative to the rapidity formulation is approximately 100×–1100×, and relative to the direct PINN it is 139×–1456×. Whether this advantage carries over to more complex Painlevé equations (III–VI) is an open question.

Additional limitations. A few limitations deserve explicit mention. The prototype $y'' = y^3$ has a closed-form Jacobi-elliptic solution ($w = \mathrm{cn}(t \mid 1/2)$, $t^* = \Gamma(1/4)^2/(4\sqrt{\pi}) \approx 1.854$), so it isolates the mechanism rather than stress-testing it. There is also no convergence theorem:

Table 6: Generalization of a PINN trained at $\alpha = 1$ and evaluated at nearby parameter values without retraining.

Test α	t^*	$E_{\text{rel,near}}$	Status
0.90	1.1665	1.77×10^{-1}	degraded
1.00	1.1577	2.39×10^{-4}	baseline
1.10	1.1492	1.29×10^{-1}	degraded

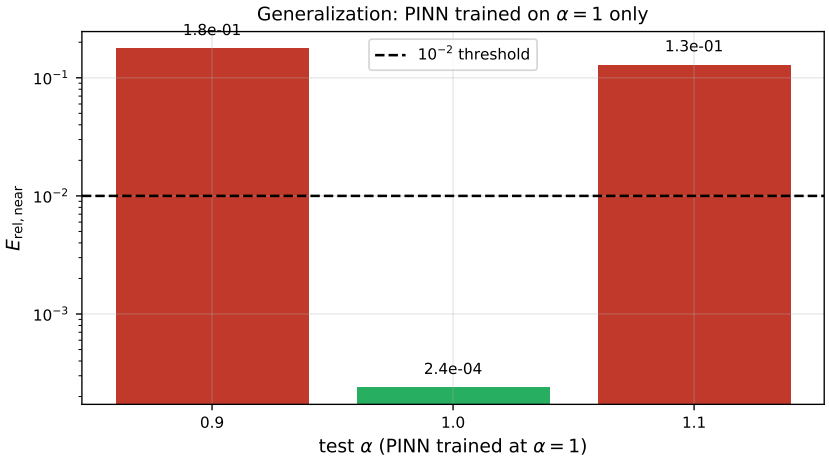


Figure 6: Generalization failure of a PINN trained at $\alpha = 1$.

boundedness of the polynomial residual does not by itself imply optimization convergence or solution accuracy. We have not benchmarked against adaptive-collocation Wu et al. (2023); Luo et al. (2026), causal-training Wang et al. (2024); Chen et al. (2026), or PINN-pathology Krishnapriyan et al. (2021); Chen et al. (2025) methods, nor against Padé-based pole-field solvers Fornberg & Weideman (2011); the comparison here is limited to the direct and rapidity formulations. Finally, boundedness of the transformed polynomial residual is not the same as well-posedness of the degenerate second-order ODE at $w = 0$, and since training stops at $0.95\,t^*$, that point is never directly evaluated.

6 Conclusion

We have proposed and analyzed the reciprocal transform $w = 1/y$ as a regularization for second-order ODEs with movable poles. For the prototype $y'' = y^3$ and Painlevé II, we proved that the transformed equations are polynomial in w and remain bounded at the pole, and we verified numerically with PINNs that the near-pole relative error decreases by median factors of 139×–523× relative to the direct formulation. An ablation separating the change of variable from residual reweighting shows that the transformation alone and the reweighting alone both fail, and only their combination reproduces the reported reciprocal-PINN improvement. Full seed variability and a direct comparison with DOP853 are reported; the PINN is not competitive as a single-solve solver, and this is stated explicitly. Extension to Painlevé III–VI, larger networks, and GPU batch inference are obvious engineering next steps.

Data Availability

All scripts, trained model checkpoints, and numerical data required to reproduce the results in this paper are available in a public GitHub repository at `https://github.com/mrmlili/`

`pinn-reciprocal-painleve`.

The repository is archived on Zenodo (DOI: 10.5281/zenodo.23814431; release v1.0.0: 10.5281/zenodo.23814432). It includes all seed-level `.npz` data, M1 ablation log, P-II multi-seed log, hardware/software requirements, and figure-generation scripts.

References

Baty, H. (2023). *Solving higher-order Lane–Emden–Fowler type equations using physics-informed neural networks: Benchmark tests comparing soft and hard constraints* (arXiv:2307.07302). arXiv. https://arxiv.org/abs/2307.07302

Butcher, J. C. (2016). *Numerical methods for ordinary differential equations* (3rd ed.). Wiley.

Cayuso, J., Herrero-Valea, M., & Barausse, E. (2025). Deep learning solutions to singular ordinary differential equations: From special functions to spherical accretion. *Physical Review D*, *111*, 064082. https://doi.org/10.1103/PhysRevD.111.064082

Chang, T.-Y., Gie, G.-M., Hong, Y., & Jung, C.-Y. (2024). *Singular layer PINN methods for Burgers' equation* (arXiv:2410.09723). arXiv. https://arxiv.org/abs/2410.09723

Chen, C., Li, X., Zhang, Y., et al. (2025). FP64 is all you need: Rethinking failure modes in physics-informed neural networks. In *Advances in Neural Information Processing Systems* (Vol. 38).

Chen, L., Wang, Y., Zhang, M., et al. (2026). Gaussian causal physics-informed neural networks. In *Proceedings of the 2025 3rd International Conference on Mathematics and Machine Learning (ICMML '25)*. https://doi.org/10.1145/3783779.3783812

Clarkson, P. A. (2003). Painlevé equations—Nonlinear special functions. *Journal of Computational and Applied Mathematics*, *153*(1–2), 127–140.

Conte, R., & Musette, M. (2008). *The Painlevé handbook*. Springer.

Cuomo, S., Schiano Di Cola, V., Giampaolo, F., Rozza, G., Raissi, M., & Piccialli, F. (2022). Scientific machine learning through physics-informed neural networks: Where we are and what's next. *Journal of Scientific Computing*, *92*, 88. https://doi.org/10.1007/s10915-022-01939-z

Deift, P., & Zhou, X. (1995). Asymptotics for the Painlevé II equation. *Communications on Pure and Applied Mathematics*, *48*(3), 277–337.

Faisal, S., Sabir, Z., Khan, S. U., Aamir, M., & Cyran, K. A. (2025). A novel heuristic Morlet wavelet neural network design for the Painlevé equation-II arising in nonlinear optics. *Scientific Reports*, *15*, 24713. https://doi.org/10.1038/s41598-025-10096-w

Fokas, A. S., Its, A. R., Kapaev, A. A., & Novokshenov, V. Yu. (2006). *Painlevé transcendents: The Riemann–Hilbert approach*. American Mathematical Society.

Fornberg, B., & Weideman, J. A. C. (2011). A numerical methodology for the Painlevé equations. *Journal of Computational Physics*, *230*(15), 5957–5973. https://doi.org/10.1016/j.jcp.2011.04.007

Gromak, V. I., Laine, I., & Shimomura, S. (2002). *Painlevé differential equations in the complex plane*. de Gruyter.

Hairer, E., Norsett, S. P., & Wanner, G. (1993). *Solving ordinary differential equations I: Nonstiff problems* (2nd ed.). Springer.

Ince, E. L. (1956). *Ordinary differential equations*. Dover.

Israr, M. Z., & Khan, I. (2025). Causal physics-informed neural networks for singular and singularly perturbed boundary value problems in chemical systems. *Global Integrated Mathematics*, *1*(1), 13–27.

Karniadakis, G. E., Kevrekidis, I. G., Lu, L., Perdikaris, P., Wang, S., & Yang, L. (2021). Physics-informed machine learning. *Nature Reviews Physics*, *3*, 422–440.

Krishnapriyan, A., Gholami, A., Zhe, S., Kirby, R., & Mahoney, M. W. (2021). Characterizing possible failure modes in physics-informed neural networks. In *Advances in Neural Information Processing Systems* (Vol. 34, pp. 26548–26560).

Lagaris, I. E., Likas, A., & Fotiadis, D. I. (1998). Artificial neural networks for solving ordinary and partial differential equations. *IEEE Transactions on Neural Networks*, *9*(5), 987–1000.

Luo, F., Yang, Y., Chen, Z., Zheng, J., & Qin, T. (2026). Adaptive residual-gradient sampling PINNs for unsteady wake reconstruction around offshore cylinder arrays. *Ocean Engineering*. https://doi.org/10.1016/j.oceaneng.2026.128317

Mishra, S., & Molinaro, R. (2022). Estimates on the generalization error of physics-informed neural networks for approximating PDEs. *IMA Journal of Numerical Analysis*, *43*(1), 1–43.

Olver, F. W. J., Olde Daalhuis, A. B., Lozier, D. W., Schneider, B. I., Boisvert, R. F., Clark, C. W., Miller, B. R., Saunders, B. V., Cohl, H. S., & McClain, M. A. (Eds.). (2025). *NIST Digital Library of Mathematical Functions* (Release 1.2.4), §32.17. https://dlmf.nist.gov/32.17

Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, *378*, 686–707.

Sirignano, J., & Spiliopoulos, K. (2018). DGM: A deep learning algorithm for solving partial differential equations. *Journal of Computational Physics*, *375*, 1339–1364.

Trefethen, L. N. (2000). *Spectral methods in MATLAB*. SIAM.

Urbán, J. F., Stefanou, P., & Pons, J. A. (2025). Unveiling the optimization process of physics informed neural networks: How accurate and competitive can PINNs be? *Journal of Computational Physics*, *523*, 113656. https://doi.org/10.1016/j.jcp.2024.113656

Wang, S., Sankaran, S., & Perdikaris, P. (2024). Respecting causality is all you need for training physics-informed neural networks. *Computer Methods in Applied Mechanics and Engineering*, *421*, 116813. https://doi.org/10.1016/j.cma.2024.116813

Wu, C., Zhu, M., Tan, Q., Kartha, Y., & Lu, L. (2023). A comprehensive study of non-adaptive and residual-based adaptive sampling for physics-informed neural networks. *Computer Methods in Applied Mechanics and Engineering*, *403*, 115671. https://doi.org/10.1016/j.cma.2022.115671